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The Cost-effectiveness of Public and Private Schools of Choice in Wisconsin

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ABSTRACT

I calculate the cost-effectiveness of private schools participating in school voucher programs, independent charter schools, and district-authorized charter schools, compared to traditional public schools in Wisconsin for the 2017–18 school year. I calculate cost-effectiveness by dividing the Accountability Report Card score for each school by the public dollars invested in each school. Using a cautious analytic approach, I find that private and independent charter schools tend to be more cost-effective than traditional public schools. The most cautious estimates from the regression models controlling for school-level student demographics suggest that independent charter schools and private schools of choice are 29 percent to 30 percent more cost-effective, respectively, than traditional public schools in Wisconsin. Cost-effectiveness does not differ between district-authorized charter school and traditional public school sectors.

KEYWORDS

School choice; charter schools; school cost-effectiveness; school vouchers

Introduction

The United States invests over $660 billion for K-12 education, or over $13,000 per student, each year, on average.[1] Real education expenditures in the U.S. have nearly quadrupled in the past half century without consistent improvements in student outcomes (Hanushek, 1997, 2015a, 2015b; Hanushek & Lindseth, 2009, 2010). Because education dollars are scarce resources, and because students' academic success is important for society, it's vital to examine which education sector delivers the most "bang for the buck."

In theory, private schools and public charter schools might be more cost-effective than traditional public schools because of competitive pressures (Friedman, 1955). Economists argue that traditional public schools hold significant monopoly power because of residential assignment and funding through property taxes (Chubb & Moe, 1988, 1990; Hoxby, 2007). This monopoly power might allow traditional public school districts to spend dollars less efficiently without much bottom-up accountability. For example,

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a school without strong incentives to spend wisely might allocate more dollars toward the number of administrators and support staff rather than direct instruction (e.g., DeAngelis & Barnard, 2020; Scafidi, 2012, 2017). Private school vouchers reduce the costs associated with exiting the traditional public school system by allowing families to use public funds to pay for private school tuition and fees. Charter school laws similarly reduce monopoly power by giving families the option to attend privately managed public schools regardless of the default public school assignment.

Theoretically, it is possible that private schools and public charter schools have stronger financial incentives to spend scarce education dollars efficiently than traditional public schools because schools of choice must attract their customers (Friedman, 1955). We may also expect school choice to lead to quality enhancements without increasing costs by allowing for better matches between schools and students (DeAngelis & Holmes Erickson, 2018). Improved matches between schools and students might improve cost-effectiveness through specialization (Smith, 2010). For example, students with similar interests may require a smaller range of extracurricular activities and students with similar ability levels may require less experienced teachers for the same outcomes as schools with more diverse student bodies.

Public and private schools of choice might also need to spend education dollars more efficiently if the school funding formula systematically funds them at lower levels, or because they are less likely to have budgetary regulations than traditional public schools (Shakeel & DeAngelis, 2017). More budgetary autonomy in private and public charter schools might allow them to be more cost-effective since the flexibility could allow school leaders to use their local knowledge to spend education dollars wisely. On the other hand, more budgetary autonomy could lead to lower levels of cost-effectiveness if school leaders are not sufficiently trained on how to manage school finances correctly. In theory, public and private schools of choice may be less cost-effective than traditional public schools if they attract families by allocating substantial amounts of resources toward advertising rather than improving educational services (Lubienski, 2007). It is also possible for traditional public schools to demonstrate cost-effectiveness advantages over private schools and public charter schools because of economies of scale and because central offices could reduce per pupil overhead costs.

It's also possible that cost-effectiveness levels differ between public charter school and private school sectors. The two sectors have differing levels of regulatory burdens, public funding levels, and networks of educators. For example, because most private schools are religious, they are more likely to have access to priests and nuns who could serve as relatively well-educated teachers and administrators at generally lower costs.

This is the first study to calculate the cost-effectiveness of private schools participating in choice programs compared to traditional public schools in

Wisconsin. Cost-effectiveness is "the efficacy of a program in achieving given outcomes in relation to the program costs" (Rossi, Lipsey, & Freeman, 2003). Using per pupil funding and Accountability Report Card data[2] from the Wisconsin Department of Public Instruction, I calculate the cost-effectiveness of private schools participating in choice programs compared to traditional public schools in 26 cities for the 2017–18 school year. I calculate the cost-effectiveness of independent charter schools compared to traditional public schools for the same year in two cities: Racine and Milwaukee. Additionally, I calculate the cost-effectiveness of district-authorized charter schools compared to traditional public schools in 52 cities.

I find that private schools receive 29 percent less funding per pupil, and independent charter schools receive 22 percent less funding per pupil, than traditional public schools, on average. After controlling for differences in student populations between sectors, I find that private schools produce 2.08 more points on the Accountability Report Card for every $1,000 invested than traditional public schools, demonstrating a 30 percent cost-effectiveness advantage for private schools. Independent charter schools produce 2.27 more points on the Accountability Report Card for every $1,000 invested than traditional public schools, demonstrating a 33 percent cost-effectiveness advantage for independent charter schools. These cost-effectiveness advantages are 36 percent for private schools of choice and 29 percent for independent charter schools in a nonlinear model replacing the Accountability Report Card score with its natural log. District-authorized charter schools receive about the same amount of per pupil funding as traditional public schools, and cost-effectiveness does not differ between the two sectors.

Literature review

Several prior studies have calculated the cost-effectiveness of public charter schools and traditional public schools in the U.S. A study of public charter schools across 21 states and the District of Columbia found a 40 percent cost-effectiveness advantage relative to traditional public schools (Wolf et al., 2014). Other studies have calculated cost-effectiveness for public charter schools and traditional public schools at the city level (DeAngelis & DeGrow, 2018; DeAngelis, Wolf, Maloney, & May, 2018a; DeAngelis, Wolf, et al., 2019). DeAngelis et al. (2018a) found that public charter schools were 31 percent more cost-effective than traditional public schools across eight U.S. cities in the 2014 school year. The team's follow-up study of the same eight cities found that the public charter school advantage increased to 40 percent in the 2016 school year (DeAngelis, Wolf, et al., 2019). Although Wolf et al. (2014), DeAngelis et al. (2018a), and DeAngelis, Wolf, et al. (2019) used complete funding data (from all public and nonpublic sources), they were not able to control for differences in students across sectors when estimating differences in per pupil funding. The current study builds on each of these evaluations by including several controls for differences in students between sectors, examining school-level data, and including private schools of choice.

DeAngelis and DeGrow (2018) found that public charter schools were about 32 percent more cost-effective than traditional public schools across 71 cities in Michigan, and their results remained statistically significant after controlling for the following student characteristics: percent of female students, percent of English Language Learners, percent of student requiring special services, percent of economically disadvantaged students, and the percent of minority students. A previous study has provided a cost-benefit analysis including private schools of choice. Wolf and McShane (2013) performed a cost-benefit analysis of the D.C. Opportunity Scholarship Program using data from an experimental evaluation (Wolf et al., 2013) of the program and found an overall benefit to cost ratio of 2.62. Wolf and McShane (2013) used the experimental evaluation's finding of a 12-percentage point increase in high school graduation to estimate economic returns (in dollars) relative to program costs.

Flanders (2017) is the first study to include control variables (percent of students qualifying for the federal lunch program, percent of nonwhite students, and percent of English Language Learners) while examining the cost-effectiveness of public charter schools relative to traditional public schools at the city level in Wisconsin. Flanders (2017) found a 33 to 42 percent cost-effectiveness advantage for independent charter schools in science and math, respectively, in Milwaukee in 2014–15. However, Flanders (2017) did not find any statistically significant cost-effectiveness advantages for instrumentality charter schools at the $p < .05$ level.

The current study builds on Flanders (2017) by examining cost-effectiveness of private schools of choice, using several additional control variables, including a more holistic outcome measure (the Accountability Report Card Score), using more recent data (2017–18), and examining schools across all Wisconsin cities. The current study is also the first that I am aware of to calculate the cost-effectiveness of private schools of choice. In addition, this is the first study to examine the cost-effectiveness of four school sectors in the same evaluation (i.e. traditional public, district-authorized charter, independent charter, and private).

School choice in Wisconsin

Wisconsin is the home of the longest-standing modern voucher programs in the country – the Milwaukee Parental Choice Program – and three other private school choice programs. Wisconsin's first private school choice program launched in 1990[3] and the state's first charter school was authorized in

1994.[4] Wisconsin's four private school choice programs served over 40,000 students during the 2018–19 school year,[5] and over 43,000 students were enrolled in public charter schools in the state in the same year.[6] The Milwaukee Parental Choice Program, the Racine Private School Choice Program, and the statewide Parental Choice Program are means-tested based on family income. Students with an active Individualized Education Plan are eligible for Wisconsin's Special Needs Scholarship Program.

Evaluations of the Milwaukee Parental Choice Program find null to positive effects on test scores (Greene, Peterson, & Du, 1999; Rouse, 1998; Wolf, 2012), civic engagement (DeAngelis & Wolf, 2019b; Fleming, 2014; Fleming, Mitchell, & McNally, 2014), crime reduction (DeAngelis & Wolf, 2019a), college enrollment (Wolf, Witte, & Kisida, 2019), degree attainment (Chingos et al., 2019), and high school graduation (Cowen, Fleming, Witte, Wolf, & Kisida, 2013). However, all but two (Greene et al., 1999; Rouse, 1998) of the evaluations of the Milwaukee Parental Choice Program used non-experimental research designs.

There were 234 public charter schools operating in Wisconsin in the 2017–18 school year.[7] There are two main types of public charter schools in the state: independent charter schools and district-authorized charter schools. There were 23 independent charter schools and 211 district-authorized charter schools in the state in 2017–18. Independent charter schools have more autonomy than district-authorized charter schools in Wisconsin.[8] Independent charter school employees are not employed by any public school district but are instead employed by the operating organization of the charter school. Independent charter schools are also authorized by entities other than the public school districts such as the City of Milwaukee, the chancellor of any institution in the University of Wisconsin (UW) System, and the Office of Educational Opportunity (UW System). District-authorized charter schools include instrumentality and non-instrumentality charter schools in Wisconsin. The main difference between instrumentality and non-instrumentality charter schools is that non-instrumentality charter schools can choose their own staff, teachers, and principals. There were 177 instrumentality charter schools and 34 non-instrumentality charter schools in the state in 2017–18. According to the Wisconsin Legislative Fiscal Bureau, district-authorized charter schools received funding in a similar manner to traditional public schools in 2017–18.[9]

The robust public and private school choice sectors in Wisconsin make the state a strong candidate for analyses comparing the cost-effectiveness of different types of schools (DeAngelis & Flanders, 2019; Flanders, 2018). The state is also a strong candidate for a cost-effectiveness analysis because the Wisconsin Department of Public Instruction publishes publicly available data on school quality and student demographics for traditional public, district-authorized charter, independent charter, and private schools of choice.

Data and methods

I use per pupil funding and Accountability Report Card data from the Wisconsin Department of Public Instruction for the most recent 2017–18 school year.[10] In Wisconsin, the maximum voucher amount was $8,176 for private school students in grades 9 through 12 and $7,530 for private school students in kindergarten through 8th grade.[11] The funding amount used for each private choice school in subsequent analyses is equal to the proportion of students in grades 9 through 12 at the school times $8,176 plus the proportion of students in kindergarten through 8th grade time $7,530. For example, if half of a private school's population of students was in kindergarten through 8th grade and the other half was in grades 9 through 12, then the voucher funding amount used in the analysis would be $7,853 ($7,530 times 50 percent plus $8,176 times 50 percent). Any cost-effectiveness advantages favoring private schools found in subsequent analyses should be considered cautious because the voucher amount used is a maximum. In most cases, private schools are required to accept the voucher funding amount as payment-in-full each year. This regulation could deter more expensive private schools from accepting voucher students (DeAngelis, Burke, & Wolf, 2019). The exception to the rule is that parents of students in grades 9–12 with an income greater than 220 percent of the federal poverty level may be charged additional tuition above the voucher amount.[12]

This cautious approach applies to the findings for independent charter schools as well, as the $8,395 amount used in the analysis is the maximum public funding allowed for independent charter schools per pupil in 2017–18.[13] The funding amount used for traditional public schools and district-authorized charter schools is the state equalized aid and the local levy, or about $10,688 per child.[14] State categorical aid and federal education dollars are excluded from the total amount used in the analyses, further making any findings of private school and independent charter school sector advantages cautious in nature. Federal revenues were about $965 per pupil (about 7.5 percent of total revenues) in the 2014–15 school year.[15] Nonpublic funding is also excluded from the subsequent analyses. Because revenues might fluctuate significantly from one year to the next, the results from the 2017–18 school year could overestimate or underestimate the relative cost-effectiveness of schools in the sample. In other words, the results presented in the current study should not be extrapolated to other school years.

The Accountability Report Card scores range from 0 to 100 and include data on multiple indicators for multiple years. The four priority areas of the Accountability Report Card are student achievement, growth, closing gaps, and on-track and post-secondary success. The student achievement priority area measures English Language Arts and mathematics performance for all students taking the Forward, ACT+ Writing, and DLM exams in the Wisconsin Student Assessment System (WSAS) in grades 3 through 8 and grade 11. Scores are based on the distribution of student achievement across four WSAS performance levels (i.e. below basic, basic, proficient, and advanced) for the three most recent school years. More weight is assigned to more recent school years and years with more tested students. The Accountability Report Card also partially accounts for differences in students across schools by assigning more weight to growth in test scores for schools with higher proportions of economically disadvantaged students. The growth priority area uses data from students in grades 3 through 8 and is based on the Forward Exam. The growth score uses value-added methodology and controls for student background characteristics such as economic disadvantage, prior academic achievement, disability status, English Language Learner status, gender, and race/ethnicity.

The analyses for private school cost-effectiveness include school-level data from 118 private schools serving 29,696 students and 487 traditional public schools serving 266,682 students across 26 cities.[16] Only 26 cities were analyzed because these cities had Accountability Report Card score data for private schools of choice and traditional public schools in the 2017–18 school year. The analyses for independent charter school cost-effectiveness include school-level data from 18 independent charter schools serving 7,652 students and 152 traditional public schools serving 81,691 students across two cities: Racine and Milwaukee. Racine and Milwaukee are the only two cities included in the independent charter school analysis because independent charter schools only existed in those two locations in the state in the 2017–18 school year. The analyses for district-authorized charter school cost-effectiveness include school-level data from 109 district-authorized charter schools serving 27,715 students and 519 traditional public schools serving 271,270 students across 52 cities. Only 52 cities were analyzed because these cities had Accountability Report Card score data for district-authorized charter schools and traditional public schools in the 2017–18 school year.

Schools are dropped from the analysis if they are missing Accountability Report Card scores for the 2017–18 school year.[17] There are two main reasons why scores are not reported for some schools: (1) the school did not have at least 20 students in testable grades, or (2) the school is a high school that did not have four years of data to generate a four-year graduation rate. Independent charter schools and district-authorized charter schools are evaluated separately in this study.

Following the existing literature (DeAngelis & DeGrow, 2018; DeAngelis et al., 2018a; DeAngelis, Wolf, et al., 2019; Wolf et al., 2014), I calculate each sector's cost-effectiveness using the following equation:

$$Cost\ Effectiveness = \frac{Accountability\ Report\ Card\ Score}{Per\ Pupil\ Revenue(in\ Thousands)}$$

The outcome is the Accountability Report Card score, while the cost is the revenue allocated to each student (in thousands of U.S. dollars). Dividing the outcome of schooling by its cost generates the measure of cost-effectiveness. In Milwaukee, for example, the average student-weighted Accountability Report Card score was 59.37 for traditional public schools.[18] The average student-weighted funding amount was $10,595 for traditional public schools. In other words, traditional public schools in Milwaukee delivered 5.60 Accountability Report Card points per $1,000 of per pupil funding. On the other hand, private schools participating in a voucher program in Milwaukee received a student-weighted Accountability Report Card score of 67.91, on average. The average student-weighted funding amount was $7,600 for private schools in Milwaukee. Therefore, private schools participating in a choice program in Milwaukee delivered 8.94 Accountability Report Card points per $1,000 of per pupil funding. Private schools delivered 3.34 more Accountability Report Card points than traditional public schools, a 60 percent cost-effectiveness advantage favoring private schools in Milwaukee. These computations for Milwaukee can be found below:

Cost-Effectiveness for Milwaukee Traditional Public Schools:

$$\frac{59.37\ points}{\$10,595} = \frac{5.60\ points}{\$1,000}$$

Cost-Effectiveness for Milwaukee Choice Private Schools:

$$\frac{67.91\ points}{\$7,853} = \frac{8.94\ points}{\$1,000}$$

Milwaukee Choice Advantage (or Disadvantage):

$$\frac{8.94\ points\ per\ \$1,000\ invested\ in\ private}{5.60\ points\ per\ \$1,000\ invested\ in\ traditional\ public} - 1 = 60\ percent$$

The next sections provide results for the cost-effectiveness of Wisconsin's traditional public, district-authorized charter, independent charter, and private school sectors.

Simple results for private schools

School-level per-pupil funding amounts and Accountability Report Card scores are compared across sectors by aggregating the data up to the city-level. Overall, traditional public schools across the 26 cities receive $10,777 per pupil and private schools receive $7,853 per pupil (Table 1). In other words, private schools receive $2,906 less funding per pupil than

Table 1. Accountability report card points per thousand dollars funded, by city.

	Traditional Public Schools			Private Choice Schools			Difference (#)	Difference (%)
City	Accountability Report Card Score	Per Pupil Revenue	Points per $1,000 Funded	Accountability Report Card Score	Per Pupil Revenue	Points per $1,000 Funded	Points per $1,000 Funded	Points per $1,000 Funded
La Crosse	68.29	$11,825	5.78	80.10	$7,530	10.64	4.86	84%
Madison	72.32	$11,999	6.03	82.80	$7,530	11.00	4.97	82%
Racine	58.51	$11,688	5.01	68.85	$7,719	8.92	3.91	78%
Wauwatosa	77.91	$10,382	7.50	92.80	$7,530	12.32	4.82	64%
Milwaukee	59.37	$10,595	5.60	67.91	$7,655	8.87	3.27	58%
West Allis	68.35	$10,634	6.43	76.24	$7,592	10.04	3.62	56%
Green Bay	67.37	$10,502	6.41	74.80	$7,530	9.93	3.52	55%
Janesville	69.87	$9,882	7.07	80.90	$7,530	10.74	3.67	52%
Sheboygan	69.56	$10,570	6.58	74.50	$7,810	9.54	2.96	45%
Waupun	67.47	$9,439	7.15	80.00	$7,720	10.36	3.21	45%
Appleton	72.09	$9,930	7.26	82.20	$8,176	10.05	2.79	38%
Mequon	86.52	$10,769	8.03	82.10	$7,530	10.90	2.87	36%
Waukesha	73.65	$10,262	7.18	72.80	$7,530	9.67	2.49	35%
Brookfield	87.96	$11,678	7.53	74.30	$7,530	9.87	2.34	31%
Fond du Lac	71.70	$9,436	7.60	73.80	$7,742	9.53	1.93	25%
Sturtevant	70.28	$11,688	6.01	57.10	$7,530	7.58	1.57	26%
Elm Grove	88.12	$11,678	7.55	69.60	$7,530	9.24	1.70	22%
Burlington	75.28	$11,050	6.81	61.90	$7,530	8.22	1.41	21%
Kenosha	68.89	$10,899	6.32	58.02	$7,695	7.54	1.22	19%
South Milwaukee	69.83	$9,990	6.99	59.90	$7,530	7.95	0.96	14%
Beloit	64.14	$10,276	6.24	56.30	$7,971	7.06	0.82	13%
Whitefish Bay	90.26	$11,289	8.00	73.50	$8,176	8.99	0.99	12%
Jackson	88.20	$9,497	9.29	84.90	$8,176	10.38	1.10	12%
New Berlin	83.13	$11,704	7.10	61.10	$7,713	7.92	0.82	12%
Wausau	73.82	$10,655	6.93	53.67	$7,530	7.13	0.20	3%
Greendale	81.30	$11,298	7.20	43.70	$8,176	5.34	(1.85)	-26%
Unweighted Average	74.01	$10,754	6.91	70.91	$7,700	9.22	2.31	34%
Student-Weighted Average	68.27	$10,777	6.36	68.10	$7,667	8.67	2.32	36%

traditional public schools, representing a funding disparity of 27 percent favoring traditional public schools. This large funding gap might be a lower bound estimate of the actual disparity because the voucher funding amount used in the analysis is the maximum public funding a private school can receive per student. This overall revenue disparity finding is about the same as the 27 to 29 percent gaps found for charter schools in eight U.S. cities outside of Wisconsin (DeAngelis, Wolf, Maloney, & May, 2018b; Wolf, Maloney, May, & DeAngelis, 2017).

Traditional public schools in the 26 cities deliver 6.40 report card points for every $1,000 invested by the public. Private schools deliver 8.67 report card points for every $1,000 invested by the public. In other words, private schools produce 2.27 more report card points than traditional public schools for every $1,000 investment, representing a 36 percent cost-effectiveness advantage favoring private schools in Wisconsin. A straight across-city average – which does not weight for the number of students in each city – finds a 2.12 point, or 31 percent, advantage favoring private schools across the 26 cities.

Simple results for public charter schools

Descriptive results for independent charter schools in Milwaukee and Racine are presented in Table 2. Overall, traditional public schools across the two cities receive $10,819 per pupil and independent charter schools receive $8,395 per pupil. In other words, independent charter schools receive $2,424 less funding per pupil than traditional public schools, representing a funding disparity of 22 percent favoring traditional public schools in the

Table 2. Accountability report card points per thousand dollars funded, by city.

	Traditional Public Schools			Independent Charter Schools			Difference (#)	Difference (%)
City	Accountability Report Card Score	Per Pupil Revenue	Points per $1,000 Funded	Accountability Report Card Score	Per Pupil Revenue	Points per $1,000 Funded	Points per $1,000 Funded	Points per $1,000 Funded
Racine	58.51	$11,688	5.01	68.70	$8,395	8.18	3.18	63%
Milwaukee	59.37	$10,595	5.60	72.85	$8,395	8.68	3.07	55%
Unweighted Average	58.94	$11,141	5.30	70.77	$8,395	8.43	3.04	59%
Student Weighted Average	59.19	$10,819	5.48	72.56	$8,395	8.64	3.09	58%

study sample. This funding disparity favoring traditional public schools is near the statewide gap of 25 percent found in Wisconsin by Batdorff, Doyle, Finch, and Hassel (2010), but smaller than the 41 percent statewide funding gap reported by Batdorff et al. (2014). The difference in funding inequity estimates might be because of the difference in the year of data evaluated or because Batdorff et al. (2014) included all sources of public and nonpublic funding. The gap reported in this analysis might also be a lower bound for the true disparity because the independent charter school per pupil funding amount used is a statewide maximum.

Traditional public schools in the sample of two cities deliver 5.48 report card points for every $1,000 in per pupil funding. Charter schools deliver 8.64 report card points for every $1,000 invested by the public. In other words, charter schools produce 3.09 more report card points than traditional public schools for every $1,000 investment, representing a 58 percent cost-effectiveness advantage favoring independent charter schools in Wisconsin. A straight across-city (unweighted) average finds a 3.04 point, or 59 percent, advantage favoring charter schools across the two cities. The independent charter school cost-effectiveness advantage is 63 percent in Racine and 55 percent in Milwaukee. These cost-effectiveness advantages are larger than the 39 to 40 percent statewide cost-effectiveness results found by Wolf et al. (2014) and the 33 to 42 percent cost-effectiveness advantage found for independent charter schools in Milwaukee by Flanders (2017). The difference in findings could be explained by the difference in the outcome examined or the year of data evaluated. Flanders (2017) also included controls for differences in students between sectors, whereas these descriptive results do not include control variables.

Descriptive results for district-authorized charter schools across 52 cities are presented in Table 3. Per pupil funding levels are about the same between the two sectors, but slightly higher Accountability Report Card Scores make district-authorized charter schools about 5 percent more cost-effective than traditional public schools. However, all of the results presented in this section are merely descriptive and do not control for differences in students between school sectors. The preferred methodology and results are presented in subsequent sections.

Table 3. Accountability report card points per thousand dollars funded, by city.

	Traditional Public Schools			District-Authorized Charter Schools			Difference (#)	Difference (%)
City	Accountability Report Card Score	Per Pupil Revenue	Points per $1,000 Funded	Accountability Report Card Score	Per Pupil Revenue	Points per $1,000 Funded	Points per $1,000 Funded	Points per $1,000 Funded
Campbellsport	70.41	$10,772	6.54	85.40	$9,673	8.83	2.29	35%
Oconto	70.53	$9,444	7.47	93.30	$9,444	9.88	2.41	32%
Fitchburg	68.20	$10,842	6.29	87.20	$10,842	8.04	1.75	28%
Milwaukee	59.37	$10,595	5.60	74.36	$10,536	7.06	1.45	26%
Kaukauna	71.95	$10,495	6.86	90.10	$10,495	8.58	1.73	25%
Eagle	83.40	$10,333	8.07	91.70	$9,284	9.88	1.81	22%
Ripon	73.67	$10,200	7.22	88.61	$10,200	8.69	1.46	20%
Neenah	77.39	$9,681	7.99	92.50	$9,681	9.55	1.56	20%
Ashland	60.09	$9,873	6.09	71.70	$9,873	7.26	1.18	19%
Minocqua	77.27	$13,754	5.62	80.16	$11,532	6.95	1.33	24%
Columbus	65.94	$11,000	5.99	77.30	$11,000	7.03	1.03	17%
Merrill	65.48	$9,922	6.60	76.50	$9,922	7.71	1.11	17%
Oshkosh	72.43	$10,504	6.90	82.05	$10,504	7.81	0.92	13%
Kenosha	68.89	$10,899	6.32	78.00	$10,900	7.16	0.84	13%
Wales	79.58	$10,346	7.69	88.10	$10,346	8.52	0.82	11%
Nekoosa	67.73	$10,574	6.41	74.40	$10,574	7.04	0.63	10%
Madison	72.32	$11,999	6.03	79.40	$12,016	6.61	0.58	10%
La Crosse	68.29	$11,825	5.78	74.47	$11,825	6.30	0.52	9%
Blair	71.64	$9,327	7.68	77.90	$9,327	8.35	0.67	9%
Fond du Lac	71.70	$9,436	7.60	76.12	$9,436	8.07	0.47	6%
River Falls	79.59	$9,411	8.46	84.00	$9,411	8.93	0.47	6%
Little Chute	77.48	$9,980	7.76	81.59	$9,980	8.18	0.41	5%
Cameron	70.63	$9,788	7.22	74.30	$9,788	7.59	0.38	5%
Augusta	67.76	$12,569	5.39	70.60	$12,569	5.62	0.23	4%
Viroqua	68.51	$9,623	7.12	71.10	$9,623	7.39	0.27	4%
Sparta	70.14	$9,535	7.36	71.76	$9,535	7.53	0.17	2%
Verona	72.59	$10,940	6.64	73.42	$10,842	6.77	0.14	2%
Albany	62.55	$12,042	5.19	63.30	$12,042	5.26	0.06	1%
Weston	72.11	$10,585	6.81	72.50	$10,585	6.85	0.04	1%
Appleton	72.09	$9,930	7.26	72.54	$9,965	7.28	0.02	0%
Monona	76.74	$11,559	6.64	79.80	$12,016	6.64	0.00	0%
Hayward	71.80	$9,729	7.38	70.73	$9,729	7.27	(0.11)	-1%
Sheboygan	69.56	$10,570	6.58	68.13	$10,570	6.45	(0.13)	-2%
Lodi	75.49	$10,969	6.88	73.10	$10,969	6.66	(0.22)	-3%
Waukesha	73.65	$10,262	7.18	71.19	$10,262	6.94	(0.24)	-3%
Eau Claire	75.58	$10,423	7.25	73.00	$10,423	7.00	(0.25)	-3%
Elkhorn	74.29	$9,792	7.59	71.70	$9,792	7.32	(0.26)	-3%
Wisconsin Rapids	68.96	$10,724	6.43	65.90	$10,724	6.15	(0.29)	-4%
Hortonville	78.19	$9,578	8.16	73.40	$9,578	7.66	(0.50)	-6%
Medford	71.38	$9,260	7.71	64.70	$9,260	6.99	(0.72)	-9%
Land O' Lakes	82.60	$13,341	6.19	74.80	$13,341	5.61	(0.58)	-9%
Stevens Point	72.61	$9,695	7.49	63.90	$9,695	6.59	(0.90)	-12%
Wauwatosa	77.91	$10,382	7.50	68.37	$10,382	6.59	(0.92)	-12%
Fredonia	76.14	$10,103	7.54	66.20	$10,103	6.55	(0.98)	-13%
Wausau	73.82	$10,655	6.93	63.69	$10,655	5.98	(0.95)	-14%
Janesville	69.87	$9,882	7.07	59.10	$9,876	5.98	(1.09)	-15%
West Bend	75.41	$9,497	7.94	60.70	$9,497	6.39	(1.55)	-20%
McFarland	75.06	$10,866	6.91	56.86	$10,866	5.23	(1.67)	-24%
Chetek	80.56	$11,535	6.98	60.40	$11,535	5.24	(1.75)	-25%
Montello	77.20	$10,950	7.05	54.40	$10,950	4.97	(2.08)	-30%
Grantsburg	73.62	$9,203	8.00	50.50	$9,203	5.49	(2.51)	-31%
Middleton	84.97	$10,685	7.95	48.20	$10,685	4.51	(3.44)	-43%
Unweighted Average	72.79	$10,498	6.99	73.33	$10,420	7.09	0.11	2%
Student-Weighted Average	69.54	$10,547	6.63	72.52	$10,434	6.97	0.34	5%

Multiple regression results

I use ordinary least squares regression analyses to test whether the Accountability Report Card Score accounts for differences in students across schools. Accountability Report Card scores are predicted by school-level student demographics (percent Black or African American, percent Hispanic or Latino, percent Asian, percent American Indian or Alaskan Native, percent Native Hawaiian or Pacific Islander, percent two or more races, percent White, percent Economically Disadvantaged, percent Students with Disabilities, and percent Limited English Proficient), school level (Elementary, Elementary/Secondary, Junior High, Middle, High), enrollment (in hundreds), and city in Table 4. The model in Column I only includes student demographics. Column II adds city fixed effects, and Column III

Table 4. Predictors of accountability report card scores.

	(1) Accountability Report Card Score	(2) Accountability Report Card Score	(3) Accountability Report Card Score
Enrollment (100s)	−0.511***	−0.596***	−0.014
	(0.000)	(0.000)	(0.875)
Black (%)	6.724*	9.143*	4.111
	(0.040)	(0.018)	(0.245)
Hispanic (%)	6.839*	9.290*	4.269
	(0.037)	(0.016)	(0.227)
Asian (%)	6.980*	9.367*	4.336
	(0.033)	(0.015)	(0.220)
American Indian (%)	6.705*	8.919*	4.158
	(0.040)	(0.021)	(0.241)
Native Hawaiian (%)	6.517+	9.004*	4.160
	(0.056)	(0.028)	(0.264)
Two or More Races (%)	6.914*	9.374*	3.934
	(0.035)	(0.015)	(0.269)
White (%)	6.800*	9.280*	4.173
	(0.038)	(0.016)	(0.239)
Econ Disadvantage (%)	−0.162***	0.020	−0.045
	(0.000)	(0.536)	(0.146)
SWD (%)	−0.415***	−0.510***	−0.522***
	(0.000)	(0.000)	(0.000)
LEP (%)	−0.011	−0.083	−0.180*
	(0.815)	(0.307)	(0.014)
City	No	Yes	Yes
School Level	No	No	Yes
N	2019	2019	2019
R-Squared	0.4267	0.4913	0.5854

Results are average marginal effects from Ordinary Least Squares regression. "American Indian" is "American Indian or Alaskan Native." "Native Hawaiian" is "Native Hawaiian or Pacific Islander." "SWD" is "Students with Disabilities." "LEP" is "Limited English Proficient." All observations are weighted by student enrollment.

additionally controls for school level. Each models weights all observations by student enrollment.

Most of the independent variables are statistically insignificant in Column III, suggesting that Accountability Report Card Scores largely account for differences in student demographics across the state. However, Accountability Report Card scores do not appear to successfully control for the percent of students with disabilities or the percent of students with limited English proficiency across schools within the same city.

Because the Accountability Report Card Scores do not fully account for student differences across schools, ordinary least squares regression is used for the main set of analyses (Table 5). I employ ordinary least squares regression analyses of the form:

$$Cost - Effectiveness_i = \beta_0 + \beta_1 Private_i + \beta_2 Charter_i + \beta_3 X_i + \varepsilon_i$$

Table 5. Cost-effectiveness of public and private schools of choice in Wisconsin.

	(1)	(2)	(3)	(4)	(5)
	Cost-Effectiveness	Cost-Effectiveness	Cost-Effectiveness	Cost-Effectiveness	Cost-Effectiveness
Private	3.042***	3.096***	3.204***	3.135***	2.076***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Independent Charter	2.871***	2.884***	3.021***	2.917***	2.268***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
District-Authorized Charter	0.419*	0.460*	0.401*	0.358*	0.057
	(0.035)	(0.019)	(0.029)	(0.050)	(0.740)
Enrollment (100s)		0.021*	0.010	0.009	0.000
		(0.030)	(0.261)	(0.353)	(0.991)
Black (%)			0.126	0.178	0.274
			(0.720)	(0.606)	(0.425)
Hispanic (%)			0.135	0.188	0.291
			(0.701)	(0.587)	(0.397)
Asian (%)			0.144	0.196	0.296
			(0.682)	(0.572)	(0.389)
American Indian (%)			0.103	0.162	0.259
			(0.771)	(0.639)	(0.451)
Native Hawaiian (%)			0.095	0.164	0.315
			(0.801)	(0.657)	(0.392)
Two or More Races (%)			0.115	0.166	0.260
			(0.745)	(0.634)	(0.451)
White (%)			0.142	0.187	0.281
			(0.687)	(0.590)	(0.414)
Econ Disadvantage (%)				−0.010**	−0.001
				(0.002)	(0.637)
SWD (%)					−0.058***
					(0.000)
LEP (%)					−0.021**
					(0.004)
City	Yes	Yes	Yes	Yes	Yes
School Level	Yes	Yes	Yes	Yes	Yes
N	2030	2030	2019	2019	2019
R-Squared	0.6380	0.6403	0.6644	0.6687	0.6917

P-values in parentheses. + $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$. Results are average marginal effects from Ordinary Least Squares regression. All models include city fixed effects and control for school level. "American Indian" is "American Indian or Alaskan Native." "Native Hawaiian" is "Native Hawaiian or Pacific Islander." "SWD" is "Students with Disabilities." "LEP" is "Limited English Proficient." "Cost-Effectiveness" is the Accountability Report Card Score divided by average per pupil funding (in $1,000s). All observations are weighted by student enrollment.

Where school i's cost-effectiveness (Accountability Report Card Score divided by average per pupil funding in thousands of dollars) is the dependent variable in each model. *Private* is an indicator variable taking on the value of one if the school (i) is a private school of choice and zero otherwise and *Charter* is an indicator variable taking on the value of one if the school (i) is an independent charter school and zero otherwise. The coefficient on β_1 captures any cost-effectiveness advantage for private schools of choice relative to traditional public schools and the coefficient on β_2 captures any cost-

Table 6. Cost-effectiveness of public and private schools of choice in Wisconsin.

	(1)	(2)	(3)	(4)	(5)
	Cost-Effectiveness	Cost-Effectiveness	Cost-Effectiveness	Cost-Effectiveness	Cost-Effectiveness
Private	0.165***	0.167***	0.168***	0.168***	0.149***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Independent Charter	0.127***	0.128***	0.130***	0.129***	0.117***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
District-Authorized Charter	0.007*	0.008**	0.007*	0.007*	0.001
	(0.029)	(0.010)	(0.016)	(0.024)	(0.610)
Enrollment (100s)		0.001**	0.000*	0.000*	0.000
		(0.003)	(0.026)	(0.033)	(0.211)
Black (%)			−0.001	−0.001	0.001
			(0.853)	(0.913)	(0.894)
Hispanic (%)			−0.001	−0.001	0.001
			(0.869)	(0.930)	(0.859)
Asian (%)			−0.001	−0.001	0.001
			(0.882)	(0.942)	(0.853)
American Indian (%)			−0.002	−0.002	0.000
			(0.745)	(0.812)	(0.995)
Native Hawaiian (%)			−0.001	−0.000	0.003
			(0.921)	(0.997)	(0.724)
Two or More Races (%)			−0.001	−0.001	0.001
			(0.837)	(0.895)	(0.917)
White (%)			−0.001	−0.001	0.001
			(0.880)	(0.932)	(0.879)
Econ Disadvantage (%)				−0.000	0.000
				(0.128)	(0.482)
SWD (%)					−0.001***
					(0.000)
LEP (%)					−0.000**
					(0.003)
City	Yes	Yes	Yes	Yes	Yes
School Level	Yes	Yes	Yes	Yes	Yes
N	2030	2030	2019	2019	2019
R-Squared	0.8957	0.8969	0.9001	0.9004	0.9063

P-values in parentheses. + $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$. Results are average marginal effects from Ordinary Least Squares regression. All models include city fixed effects and control for school level. "American Indian" is "American Indian or Alaskan Native." "Native Hawaiian" is "Native Hawaiian or Pacific Islander." "SWD" is "Students with Disabilities." "LEP" is "Limited English Proficient." "Cost-Effectiveness" is the natural log of the Accountability Report Card Score divided by average per pupil funding (in $1000s). All observations are weighted by student enrollment.

effectiveness advantage for independent charter schools relative to traditional public schools. X is a vector of the following controls: school-level student demographics (percent Black or African American, percent Hispanic or Latino, percent Asian, percent American Indian or Alaskan Native, percent Native Hawaiian or Pacific Islander, percent two or more races, percent White, percent Economically Disadvantaged, percent Students with Disabilities, and percent Limited English Proficient), school level

(Elementary, Elementary/Secondary, Junior High, Middle, High), enrollment (in hundreds), and city.

All observations are weighted by student enrollment. The analytic sample includes data from 1,773 traditional public schools (serving 779,401 students), 119 district-authorized charter schools (serving 28,925 students), 18 independent charter schools (serving 7,652 students), and 122 private choice schools (serving 30,380 students) in Wisconsin. The fully specified model includes 2,019 school-level observations since it drops 13 of the 2,032 schools (0.6 percent) with missing data.

The cost-effectiveness advantages favoring private schools of choice and independent charter schools remain statistically significant at $p < .001$ across all five models; however, the advantage becomes smaller in size once all controls are included (Column V). The model including all controls indicates that private schools of choice produce around 2.08 more points than traditional public schools on the Accountability Report Card for every $1,000 investment, all else equal. The model with all controls also indicates that independent charter schools produce around 2.27 more points than traditional public schools on the Accountability Report Card for every $1,000 investment, all else equal. Relative to the overall traditional public school cost-effectiveness mean of 6.92, private schools demonstrate a 30 percent cost-effectiveness advantage and independent charter schools demonstrate a 33 percent cost-effectiveness advantage, all else equal. The fully specified model does not find that district-authorized charter schools are more or less cost-effective than traditional public schools.

These findings are similar to the 33 to 42 percent cost-effectiveness advantage for independent charter schools and null results for instrumentality charter schools found by Flanders (2017) in Milwaukee. The findings slightly differ from Flanders (2017) because the current study uses additional control variables, includes several more cities, uses a different outcome measure, and evaluates school-level data from 2017–18. All control variables are statistically insignificant except for the percent of students with disabilities and the percent of students with limited English proficiency, again indicating that Accountability Report Card scores tend to account for many differences in student demographics across schools.

It is possible that the relationship between per pupil funding and the Accountability Report Card Score is nonlinear. The linear model assumes that, all else equal, there will be constant returns to scale between per pupil funding and academic outcomes. The model in Table 6 attempts to address the possibility of nonlinearity. The linear measure of cost-effectiveness in this model is replaced by taking the natural log of the school's Accountability Score and dividing that number by the per pupil funding (in $1,000s). Relative to the sample mean (0.41), the results in Table 6 suggest that private schools are 36 percent more cost-effective than traditional public schools and independent charter schools are 29 percent more cost-effective than traditional public schools. The cost-effectiveness advantage for district-authorized charter schools is not statistically significant in the fully specified model.

Models in Tables A1 and A2 are provided as robustness checks and find similar results. A model using the natural log of cost-effectiveness as the dependent variable finds that the private school cost-effectiveness advantage is 26 percent and the independent charter school cost-effectiveness advantage is 33 percent, but does not detect a statistically significant difference in cost-effectiveness for district-authorized charter schools (Table A1). A model defining "cost-effectiveness" as the Accountability Report Card Score divided by the square root of per pupil funding also finds statistically significant advantages for private schools and independent charter schools (Table A2).

While the cost-effectiveness advantages for public and private schools of choice remain after controlling for these observable factors, student composition may differ on other measures that were not included in these models. Students may differ across sectors on other unobservable characteristics such as parent education, immigrant status, and parent motivation.

Conclusion and policy implications

Using publicly available data from the 2017–18 school year, I find that private schools participating in choice programs and independent charter schools tend to be more cost-effective than traditional public schools. The most cautious estimates from the ordinary least squares regression models suggest these cost-effectiveness advantages are 2.08 more points (30 percent more) on the Accountability Report Card per $1,000 in per pupil funding for private schools and 2.27 more points (33 percent more) on the Accountability Report Card per $1,000 in per pupil funding for independent charter schools. These cost-effectiveness advantages are 36 percent for private schools of choice and 29 percent for independent charter schools in a nonlinear model replacing the Accountability Report Card score with its natural log. However, measures of cost-effectiveness do not differ between district-authorized charter schools and traditional public schools.

This study has important limitations. Although the Accountability Report Card score attempts to account for differences in students' economic disadvantage across schools, the measure cannot perfectly control for all differences in students. In addition, while any cost-effectiveness advantages for public and private schools of choice remain after controlling for some observable factors, student composition may differ on other measures that were not included in these models. Students may differ across sectors on unobservable characteristics such as parent education, immigrant status, and parent motivation.

However, the direction of selection bias, if it exists, is unclear.[19] Fleming, Cowen, Witte, and Wolf (2015) find that students using the Milwaukee

Parental Choice Program have parents with lower incomes but more years of education than their matched peers in traditional public schools. DeAngelis and Wolf (2019b) similarly find that students using the Milwaukee Parental Choice Program have lower baseline math scores but higher baseline reading scores, and have parents with lower incomes but higher levels of education than their matched peers in traditional public schools. Chakrabarti (2013) finds that students applying to the Milwaukee Parental Choice Program have higher test scores and have parents with more years of education than the eligible population. However, the present study compares all students in choice schools to the entire public school population, not just the population of students that are eligible for school vouchers. Private school choice programs in the state tend to serve the least advantaged children, as 74 percent of voucher students across Wisconsin come from households earning at or below 185 percent of the federal poverty line.[20] Another limitation of the present study is that the results are based on Accountability Report Card scores. These scores likely do not capture all student outcomes that are valued by students, their families, and taxpayers. It is possible that private and charter schools have more, less, or about the same cost-effectiveness in terms of other unmeasured, yet valuable, outcomes.

This study suggests Wisconsin's private schools and independent charter schools tend to be more cost-effective than traditional public schools. More student-centered funding (allowing a higher proportion of education dollars to follow the child) might give Wisconsin's traditional public schools stronger financial incentives to become more efficient (Friedman, 1955). In addition, it's possible that giving traditional public school principals more autonomy regarding budgeting and personnel decisions could similarly lead to efficiency improvements (Chubb & Moe, 1990; Hanushek, Link, & Woessmann, 2013; Ouchi, 2006). In theory, allowing 100 percent of school funding to follow the student could also lead to more competitive pressures for traditional public schools to improve (Chakrabarti, 2008; Egalite, 2013; Hoxby, 2000; Jabbar et al., 2019).

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Appendix

Table A1. Cost-effectiveness of public and private schools of choice in Wisconsin.

	(1)	(2)	(3)	(4)	(5)
	Cost-Effectiveness	Cost-Effectiveness	Cost-Effectiveness	Cost-Effectiveness	Cost-Effectiveness
Private	0.441***	0.453***	0.471***	0.460***	0.259***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Independent Charter	0.439***	0.442***	0.467***	0.449***	0.327***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
District-Authorized Charter		0.066*	0.075*	0.066*	0.058+
		(0.045)	(0.020)	(0.031)	(0.055)
Enrollment (100s)		0.005**	0.003+	0.003	0.001
		(0.006)	(0.073)	(0.108)	(0.517)
Black (%)			0.008	0.017	0.035
			(0.894)	(0.769)	(0.534)
Hispanic (%)			0.009	0.018	0.038
			(0.874)	(0.748)	(0.500)
Asian (%)			0.011	0.020	0.039
			(0.854)	(0.730)	(0.492)
American Indian (%)			0.005	0.015	0.034
			(0.931)	(0.787)	(0.548)
Native Hawaiian (%)			0.005	0.017	0.046
			(0.938)	(0.780)	(0.443)
Two or More Races (%)			0.006	0.015	0.033
			(0.914)	(0.792)	(0.559)
White (%)			0.010	0.018	0.036
			(0.859)	(0.751)	(0.522)
Econ Disadvantage (%)				−0.002**	−0.000
				(0.001)	(0.841)
SWD (%)					−0.011***
					(0.000)
LEP (%)					−0.004**
					(0.009)
City	Yes	Yes	Yes	Yes	Yes
School Level	Yes	Yes	Yes	Yes	Yes
N	2030	2030	2019	2019	2019
R-Squared	0.5969	0.6012	0.6277	0.6328	0.6651

P-values in parentheses. + $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$. Results are average marginal effects from Ordinary Least Squares regression. All models include city fixed effects and control for school level. "American Indian" is "American Indian or Alaskan Native." "Native Hawaiian" is "Native Hawaiian or Pacific Islander." "SWD" is "Students with Disabilities." "LEP" is "Limited English Proficient." "Cost-Effectiveness" is the Accountability Report Card Score divided by average per pupil funding (in $1000s). The dependent variable is the log of cost-effectiveness. All observations are weighted by student enrollment.

Table A2. Cost-effectiveness of public and private schools of choice in Wisconsin.

	(1)	(2)	(3)	(4)	(5)
	Cost-Effectiveness	Cost-Effectiveness	Cost-Effectiveness	Cost-Effectiveness	Cost-Effectiveness
Private	0.173***	0.178***	0.190***	0.182***	0.074**
	(0.000)	(0.000)	(0.000)	(0.000)	(0.004)
Independent Charter	0.195***	0.197***	0.211***	0.200***	0.134***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
District-Authorized Charter	0.042*	0.046*	0.040*	0.035+	0.004
	(0.038)	(0.022)	(0.033)	(0.058)	(0.812)
Enrollment (100s)		0.002*	0.001	0.001	−0.000
		(0.045)	(0.354)	(0.477)	(0.842)
Black (%)			0.016	0.022	0.032
			(0.646)	(0.525)	(0.353)
Hispanic (%)			0.017	0.023	0.033
			(0.628)	(0.508)	(0.329)
Asian (%)			0.018	0.024	0.034
			(0.609)	(0.493)	(0.321)
American Indian (%)			0.015	0.021	0.031
			(0.679)	(0.540)	(0.363)
Native Hawaiian (%)			0.012	0.019	0.035
			(0.754)	(0.598)	(0.337)
Two or More Races (%)			0.015	0.021	0.030
			(0.673)	(0.555)	(0.379)
White (%)			0.018	0.023	0.032
			(0.614)	(0.511)	(0.344)
Econ Disadvantage (%)				−0.001***	−0.000
				(0.001)	(0.460)
SWD (%)					−0.006***
					(0.000)
LEP (%)					−0.002**
					(0.007)
City	Yes	Yes	Yes	Yes	Yes
School Level	Yes	Yes	Yes	Yes	Yes
N	2030	2030	2019	2019	2019
R-Squared	0.5453	0.5478	0.5819	0.5885	0.6187

P-values in parentheses. + $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$. Results are average marginal effects from Ordinary Least Squares regression. All models include city fixed effects and control for school level. "American Indian" is "American Indian or Alaskan Native." "Native Hawaiian" is "Native Hawaiian or Pacific Islander." "SWD" is "Students with Disabilities." "LEP" is "Limited English Proficient." "Cost-Effectiveness" is the Accountability Report Card Score divided by the square root of average per pupil funding (in $1000s). All observations are weighted by student enrollment.